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SATHYABAMA INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CONTINUOUS ASSESSMENT EXAM – 1(SET-A)

Programme: B.E-CSE,CSE(SPECL)/B.TECH-IT — Max. Marks: 50
Course Code: S11BLH21 — Duration: 2 Hours
Course Name: PROGRAMMING IN PYTHON — Sem : II
Batch: 2025-2029 — Date :20.2.2026

Answer All (5 × 10 = 50)

Q. No.	Questions	CO
1.	A text –based adventure game begins at the entrance of an abandoned castle. The player is given three possible actions: entering the castle, exploring the castle, or leaving the area. Each action leads to different outcomes, such as encountering a ghost, finding a magic key, ending the game peacefully. Based on the player's choice, appropriate decisions must be made using conditional statements.	co1
	or	
2	A restaurant management system needs to record menu details. The system should collect the food item name, its price, and whether it is vegetarian. Using Python built-in functions, the program must verify input data types, ensure the price is numeric, and display message based on the food category.	co1
3.	Imagine you are building a fitness app. The app wants to calculate the BMI for multiple users. Your tak is to: 1. Accept the number of users. 2. Input each user's weight and height. 3. Calculate BMI and classify them as Underweight, Normal weight, Overweight, or Obese.	co1
	Or	
4	An organization wants to strengthen its password policy. As part of the validation process, a program must analyze a given text or password and count uppercase letters, lowercase letters, digits, and special characters to ensure password strength.	co1

5.	A local electricity board requires a billing system to compute electricity charges based on the number of units consumed. The system should apply different tariff rates for different unit ranges and add a surcharge of 15%, if the bill amount exceeds Rs.2000.The billing criteria are as follows: Units consumed Rate per unit(Rs.) 0-100 5 101-200 7 201-300 10 Above 300 15	co1
	or	
6	A school plans to digitize student academic records. To manage student information efficiently, an object-oriented approach is required where student details and marks are stored, retrieved, and analyzed using a class-based design.	co2

7.	A community club maintains member records that include name and age. For security reasons, the age attribute should not be accessed directly. Instead, controlled access must be provided using getter and setter methods with proper validation.	co2
	or	
8	A student registration module is required for a school system. Each student record should be initialized with basic details and marks using a parameterized constructor, and methods should be provided to display information and calculate average marks.	co2

9.	Develop a transport tracking system that manages vehicles and their details. All vehicles have a common attribute(brand, model, year, mileage) and bike is a special type of vehicle with additional attributes such as type(sports, cruiser, scooter)and engine capacity. The system should allow: 1. Displaying vehicle details. 2. Displaying bike-specific details. 3. Updating mileage when the vehicle is used.	co2
	or	
10	A graphics application supports multiple shapes such as circles, rectangles, and triangles. Each shape has its own way of being drawn. Using a common interface, the application should be able to call the draw method for different shapes at runtime.	co2